

◆ 16.11.X Global Closure and Unified System Theorem (Final — Hard Closure)

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
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16.11.X.1 Unified State Space

$$\mathcal{S}_k = (\Psi_k, P_k, PAC_k, D_k, u_k)$$

Under the assumptions:

- $\Psi_k \in \Omega \subset \mathbb{R}^d$, Ω compact
- $P_k \geq \epsilon > 0$
- $PAC_k \in (0,1)$
- D_k bounded
- $u_k \in \mathcal{U}_{\{\mathrm{feasible}\}}$

Thus:

$$\mathcal{S}_k \in \mathcal{X}_{\{\mathrm{safe}\}}$$

16.11.X.2 Global Evolution Law

$$\mathcal{S}_{k+1} = F_{\text{global}}(\mathcal{S}_k)$$

with:

$$F_{\text{global}} = F_{\text{dyn}} \circ F_{\text{prob}} \circ F_{\text{PAC}} \circ F_{\text{eval}} \circ F_{\text{decision}} \circ F_{\text{control}}$$

Thus:

the system evolves through a fully integrated composition of all subsystems.

16.11.X.3 Closed-Loop Structure

$$\Psi \rightarrow P \rightarrow \text{PAC} \rightarrow D \rightarrow u \rightarrow \Psi$$

Thus:

the system is fully closed-loop with no external dependency.

16.11.X.4 Well-Posedness

From Sections 16.11.2 and 16.11.8:

- $F_{\{\mathrm{global}\}}$ is Lipschitz
- inputs are bounded

$\Rightarrow \exists! \mathcal{S}_k$

Thus:

the system admits a unique trajectory.

16.11.X.5 Global Lyapunov Stability

$$\mathcal{L}(\mathcal{S}) = \mathcal{E}(\Psi) + KL(P \parallel P_0)$$

$$\begin{aligned} \dot{\mathcal{L}} \\ \leq \\ -c\|\Psi\|^2 + C\sigma^2 \end{aligned}$$

Thus:

$\mathcal{L}(k)$ is non-increasing up to stochastic perturbations.

$\Rightarrow$ $\text{global stability holds}$

16.11.X.6 Robustness

Under bounded noise, control, and probability:

$$\mathbb{E}[\|\mathcal{S}_k\|^2] \leq C$$

Thus:

the system is globally bounded (ISS-type robustness).

16.11.X.7 Convergence

$$\limsup_{k \rightarrow \infty} \mathcal{S}_k = \mathcal{S}^*$$

and in the deterministic limit:

$$\sigma \rightarrow 0; \Rightarrow \mathcal{S}_k \rightarrow \mathcal{S}^*$$

Thus:

the system converges to a steady-state solution.

16.11.X.8 Optimality

$u_k =$

$\arg\max_{u \in \mathcal{U}_{\text{feasible}}} D_u(k)$

Thus:

the system implements an optimal policy.

16.11.X.9 Adaptivity

- PAC evolves dynamically
- weights adapt
- control law updates

Thus:

the system is self-adaptive.

16.11.X.10 Real-Time Closure

$$\begin{aligned} T_{\mathrm{total}} = & \\ T_{\mathrm{dyn}} + T_{\mathrm{eval}} + & \\ T_{\mathrm{decision}} + T_{\mathrm{control}} & \\ \leq T_{\mathrm{critical}} & \end{aligned}$$

Thus:

real-time execution is guaranteed.

16.11.X.11 Safety Invariance

$$\begin{aligned} \mathcal{S}_k \in \mathcal{X}_{\mathrm{safe}} \\ \Rightarrow \\ \mathcal{S}_{k+1} \in \mathcal{X}_{\mathrm{safe}} \end{aligned}$$

Thus:

the system never exits the safe domain.

16.11.X.12 Compatibility Closure

All components:

- dynamics
- PAC
- evaluation
- decision
- control

are mutually consistent within $F_{\{\mathrm{global}\}}$.

Thus:

no internal contradiction exists in the system architecture.

 Final Master Statement (Theorem-Level Closure)

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\boxed{
\text{SEKHEM-TSDD defines a fully closed, PAC-
based stochastic decision–control system that is
stable, robust, optimal, adaptive, and physically
realizable.}
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